Attack Detection and Reconstruction for CPSs via an Improved T-N-L Observer and Zonotopic Analysis

Ben Niu, Chaojiang Liang, Zhihua Guo, Xinjun Wang, *Member, IEEE*, and Xudong Zhao

***Abstract*—Safety-critical Cyber-Physical Systems (CPSs) are inherently vulnerable to malicious attacks. While detecting these attacks is crucial, handling practical unknown but bounded (UBB) disturbances without triggering conservative thresholds remains a significant challenge. To fill this gap, this paper investigates attack detection, reconstruction, and isolation for safety-critical CPSs under UBB disturbances. In order to represent complex nonlinear systems, the Takagi-Sugeno (T-S) fuzzy model is utilized. First, an improved T-N-L observer integrated with the H_∞ technique is designed to estimate the system states. To improve estimation accuracy, we develop a decoupling approach that decouples the partial disturbance matrix, effectively attenuating the effect of disturbances on the error system. Second, compared to conventional observers, a strictly tighter interval estimation of the system state is derived through the reachability analysis of the error system. Based on this, an improved attack detection method using residual reachability analysis is proposed to bypass conservative thresholds and reduce computational complexity. Furthermore, the impact of disturbances and the error system on attack reconstruction and isolation is investigated. Finally, simulation results confirm the superiority of the proposed framework. In particular, the proposed method reduces the missed detection rate to 0%, compared with 4.84% for traditional methods, and significantly improves the reconstruction accuracy for both actuator and sensor attacks. For sensor attacks, it achieves an approximate 90.1% reduction in root mean square error (RMSE), while enabling zero-false-alarm isolation for T-S fuzzy systems.**

***Index Terms*—Attack detection and reconstruction, cyber-physical systems, H_∞ technique, T-N-L observer, reachability analysis.**

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I. Introduction

RECENTLY, the cyber-physical systems (CPSs) are defined as physical systems integrated with computing and communication capabilities, encompassing many nonlinear systems. Takagi-Sugeno (T-S) fuzzy systems have emerged as a powerful framework for representing complex nonlinear systems in CPSs, including uncrewed aerial vehicle (UAV) [1], autonomous underwater vehicles [2], and multi-agent systems [3]. The inherent networked nature of these CPSs exposes them to malicious cyber attacks that threaten operational security and reliability. Any attack on the safety-critical systems can lead to economic losses or casualties. Consequently, it is critical to investigate attack detection and/or reconstruction method to detect malicious attacks.

For the purpose of improving the safety of nonlinear systems, it is crucial to investigate effective methods for malicious attack detection. In the existing literature, a zonotopic observer-based method was proposed in [4] for state estimation and fault detection of CPSs under denial-of-service (DoS) attacks. The study in [5] investigated an unknown-input-observer-based method with reachability analysis for attack detection and reconstruction in discrete-time linear CPSs subject to unknown but bounded (UBB) disturbances. A coding-based distributed filtering method for stochastic CPSs over unreliable wireless sensor networks under DoS-induced packet losses was presented in [6]. The work in [7] proposed a zonotopic observer-based method for attack detection and active attack defense in CPSs under both actuator and sensor false data injection (FDI) attacks. However, it is important to note that many existing attack detection approaches are primarily applicable to systems characterized by Gaussian-distributed noise. In practical applications, it is often more suitable to consider disturbances that are UBB. Therefore, how to describe UBB disturbances is a significant research topic. Interval observers (IOs) have also been widely studied for uncertain systems with bounded disturbances, including general IO design methods [8], interval state estimation for LPV systems [9], Luenberger-like IOs for uncertain second-order systems [10], and recent LMI-based multi-step IOs [11]. Compared with IOs, which mainly provide componentwise lower and upper bounds, zonotopic representations can preserve part

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of the coupling information among uncertain variables through generator matrices, which is beneficial for reachability-based residual analysis and attack reconstruction. As pointed out in [12], zonotopes offer superior accuracy and more compact representations than ellipsoidal sets, while maintaining lower complexity compared with polytopes. Hence, zonotopes are utilized to represent the UBB noise in this work. The study in [13] investigated attack detection in T–S fuzzy systems with UBB noise using set-membership estimation and a multi-mode high-order matrix method. Clearly, research focusing on attack detection in T–S fuzzy systems with UBB disturbances remains limited. Therefore, one primary motivation of this paper is to investigate an effective approach for detecting attacks in T–S fuzzy systems subject to UBB disturbances.

Meanwhile, the attack reconstruction presents a considerable challenge, which prompted numerous researchers to explore this field [14], [15], [16], [17]. The work [14] investigated the existence conditions of joint stealthy attacks on multisensor state estimation systems under round-robin protocol with actuator and sensor attacks. A descriptor switched sliding mode observer method was shown in [15] for addressing the secure reconstruction problem of linear CPSs under simultaneous sparse actuator and sensor attacks. A zonotopic observer-based switching control method was investigated in [16] for addressing secure estimation, attack reconstruction, and isolation in CPSs under malicious attacks. The article [17] proposed an adaptive neural-network-based event-triggered secure formation control method for nonholonomic mobile robots subject to deception attacks. However, the problem of attack reconstruction in T-S fuzzy systems has not yet been investigated. Currently, only fault reconstruction in T-S fuzzy systems has been proposed in [18] and [19]. Therefore, it is crucial to develop attack reconstruction methods in T-S fuzzy systems that balance high accuracy with computational efficiency.

Furthermore, in modern networked CPSs, malicious attacks may degrade not only system safety but also communication and energy efficiency. For example, secure control of nonlinear CPSs under DoS attacks has been investigated in sensor networks [20], while remote state estimation over two-hop CPS networks has been studied from the viewpoint of DoS attack power allocation [21]. In addition, adaptive prescribed-performance secure consensus control, predefined-time secure formation control, self-triggered secure control under deception attacks, and energy-aware transmission scheduling have also been explored for green networked CPSs [22], [23], [24], [25], [26]. These studies indicate that security, communication reliability, and resource efficiency are closely coupled in modern CPSs. In this context, efficient attack detection, reconstruction, and isolation can provide diagnostic information for identifying compromised channels and supporting subsequent attack-tolerant control or compensation, thereby reducing unnecessary resource consumption and enhancing the resilient operation of networked CPSs.

According to the above discussions, motivated by [16], [27], and [28], this paper proposes a comprehensive framework for secure state estimation, attack detection, and reconstruction in CPSs modeled by T-S fuzzy systems subject to UBB disturbances. Specifically, an improved T-N-L observer integrated with a disturbance decoupling mechanism and the H_∞ technique is designed to explicitly bound the estimation errors while attenuating disturbance effects. By leveraging zonotope-based reachability analysis, dynamic interval bounds are calculated for the system residuals to achieve non-conservative attack detection. Subsequently, the attacks are reconstructed and successfully isolated. The system architecture under malicious attacks is shown in Fig. 1.

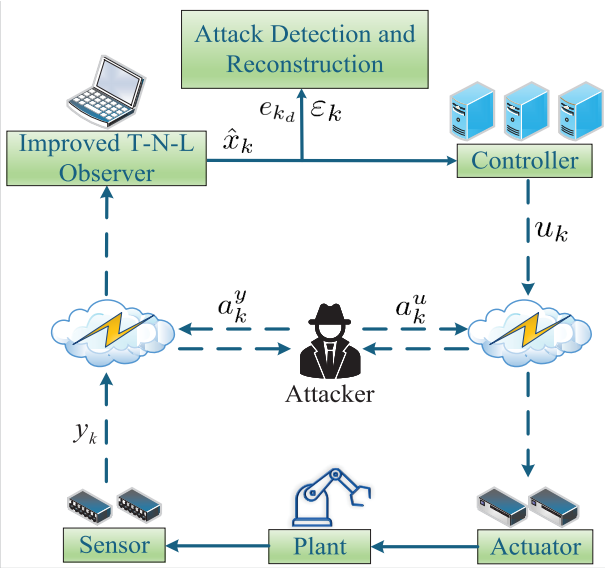


Fig. 1. The system architecture under malicious attacks.

The primary contributions of this work are summarized below:

1) To enhance state estimation accuracy in T-S fuzzy systems, an improved T-N-L observer combined with a decoupling approach is proposed. Compared with the traditional T-N-L observer without decoupling method [29], [30], tighter interval estimation of the system state can be obtained through reachability analysis. Additionally, the stability of the error system is analyzed based on linear matrix inequality.

2) An improved attack detection method based on the reachability analysis of the residue method is proposed to tackle the problem of conservative attack detection thresholds for T-S fuzzy system. Different from the methods proposed in [31] and [32], the proposed approach effectively improves the attack detection rate, reducing the missed detection rate (MDR) to 0% in simulation.

3) This work analyzes the impact of error system and disturbances on attack reconstruction and isolation for T-S fuzzy system, provides bounds on reconstruction errors. Our approach achieves more accurate attack reconstruction for both actuator and sensor attacks, notably achieving an approximate 90.1% RMSE reduction for sensor attacks, compared to the methods proposed in [16] and [33].

To enhance readability, the essential mathematical notations and core system variables employed in this paper are defined in Table I.

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TABLE I
NOTATIONS AND SYSTEM VARIABLES

Notation	Description
Mathematical Operators	
$\mathbb{R}^n$	The n-dimensional Euclidean space
$\mathbb{R}^{m\times n}$	The space of $m \times n$ real matrices
I_n	The $n \times n$ identity matrix
A^T	The transpose of matrix A
$A^\dagger$	The pseudo-inverse of matrix A
$P > 0\,(P < 0)$	Matrix P is a positive (negative) definite matrix
$\|A\|_2$	The 2-norm of matrix A
$\oplus$	The Minkowski sum
$\langle p_z, H_z\rangle$	The center and generating matrix of a zonotope
$Box(\cdot)$	The interval hull of a set
System Variables and Matrices	
x_k, u_k, y_k	System state, control input, and system output
w_k	Unknown but bounded (UBB) disturbance
w_k^a, w_k^b	Decoupled disturbance components
a_k^u, a_k^y	Actuator attacks and sensor attacks
$\hat{x}_k$	State estimation of the T-N-L observer
e_k	Error system
e_{k_d}	The error system after decoupling process
ϵ_k	System residual
A_i, B_i, C, D	System matrix, input matrix, and output matrices
E_i, F_i	Disturbance matrix and actuator attack matrix
$\bar{E}_a, \bar{E}_b(\sigma_k), M(\sigma_k)$	Parameterized matrices for disturbance decoupling
$\sigma_i(\iota_k)$	Fuzzy weights for the i-th subsystem

The rest of this paper is organized as follows: Preliminaries and system description are presented in Section II. Section III presents the main results in this paper. In Section IV, the simulation results are shown to verify the efficiency of the proposed method. The limitations of this work are shown in Section V. Section VI concludes the paper.

II. PRELIMINARIES AND SYSTEM DESCRIPTION

In this section, some basic concepts about the zonotope and the construction of the system model are presented.

Definition 1 ([34]): An interval vector $\mathbb{I} \in \mathbb{R}^n$ also known as a box, is represented as

$$\mathbb{I} = [p, q] = \{x : x \in \mathbb{R}^n, p_i \le x_i \le q_i\},$$

where $p_i = [p_1, \ldots, p_n]^T$ and $q_i = [q_1, \ldots, q_n]^T$.

Definition 2 ([34]): An i-order zonotope $\mathbb{Z} \subseteq \mathbb{R}^n$ is defined as

$$\mathbb{Z} = \langle p_z, H_z\rangle = \langle p_z + H_z z : z \in \mathbf{B}^i\rangle,$$

where $\mathbb{Z}$ is the affine transformation of the hypercube $\mathbf{B}^i = [-1, 1]^i$, $p_z \in \mathbb{R}^n$ is the center of the zonotope $\mathbb{Z}$ and $H_z \in \mathbb{R}^{n\times m}$ is known as the generating matrix of the zonotope $\mathbb{Z}$.

Definition 3 ([35]): For a set $\mathbf{G} \subseteq \mathbb{R}^n$, its interval hull is defined as the smallest interval vector enclosing it, which is represented as

$$\mathbf{G} \subseteq \text{Box}(\mathbf{G}) = [c, d], \tag{1}$$

where $c = [c_1, \ldots, c_n]^T$ and $d = [d_1, \ldots, d_n]^T$.

Definition 4 ([36]): The Minkowski sum of two sets $\mathbf{Y}$ and $\mathbf{Z}$ is defined as

$$\mathbf{Y} \oplus \mathbf{Z} = \{x + z : x \in \mathbf{Y}, z \in \mathbf{Z}\},$$

where $\oplus$ denotes the Minkowski sum.

Property 1 [36]: For zonotopes, the following equations can be obtained as

$$\mathcal{Z} = \mathcal{Z}_i \oplus \mathcal{Z}_j = \quad (p_i + p_j) \oplus [H_i \quad H_j]\mathbf{B}^{i+j}, \tag{2}$$

$$L_1\mathcal{Z} = L_1\langle p, H\rangle = \quad L_1 \odot \langle p, H\rangle = \langle L_1 p, L_1 H\rangle, \tag{3}$$

where symbol $\odot$ denotes linear map.

Property 2 [37]: The linear matrix inequality satisfies the following condition:

$$\begin{pmatrix} X & Z \\ Z^T & Y \end{pmatrix} > 0, \tag{4}$$

where X and Y are symmetric matrices, i.e, $Y = Y^T$ and $X = X^T$. Then, (4) is equivalent to

$$Y > 0, \quad X - ZY^{-1}Z^T > 0. \tag{5}$$

Lemma 1 ([38]): For matrices $X \in \mathbb{R}^{a\times b}$, $Y \in \mathbb{R}^{b\times c}$, and $Z \in \mathbb{R}^{a\times c}$, if Y has full column rank i.e., rank$(Y) = c$, the general solution of the matrix equation $XY = Z$ can be obtained as

$$X = ZY^\dagger + S(I - YY^\dagger), \tag{6}$$

where $S \in \mathbb{R}^{a\times b}$ is an arbitrary matrix.

Property 3 [39]: For the aim of limiting the order of a zonotope $\mathcal{Z} = \langle \bar{p}_k, \bar{H}_k\rangle$, the reduction operator is represented as

$$\langle \bar{p}_k, \bar{H}_k\rangle \in \langle \bar{p}_k, \hat{H}_k\rangle,$$

where $\bar{H}_k \in \mathbb{R}^{n\times p}, \hat{H}_k = \downarrow_{q,W} (\bar{H}_k)$ and $q \ge n$ denotes the maximum number of columns of matrix $\hat{H}_k = \downarrow_{q,W} (\bar{H}_k)$. First, the columns of $\bar{H}_k$ are ranked by decreasing weighted vector norm, i.e., $\bar{H}_k = [\bar{h}_k^1, \ldots, \bar{h}_k^j, \ldots, \bar{h}_k^p]$ with $\|\bar{h}_k^p\|_W^2 \ge |\bar{h}_k^{p+1}|_W^2$. Then, the set produced by the $p - q + n$ smaller columns is enclosed into a box, i.e., $rs((\bar{H}_k)_<)$, where $(\bar{H}_k)_< = [\bar{h}_k^{q-n+1}, \ldots, \bar{h}_k^p]$ and $rs((\bar{H}_k)_<) = \text{diag}(|(\bar{H}_k)_<|\mathbf{1})$. Thus, we have $\downarrow_{q,W} (\bar{H}_k) = [(\bar{H}_k)_>, (\bar{H}_k)_<] \in \mathbb{R}^{n\times q}$ with $(\bar{H}_>) = [\bar{h}_k^1, \ldots, \bar{h}_k^{q-n}]$. If $W = I$, then we represent $\downarrow_{q,W} (\bar{H}_k) = \downarrow_q (\bar{H}_k)$.

Property 4 [29]: For a series of sets $S_i \subset \mathbb{R}^n$, $i = 1, \ldots, m$, the following equation holds

$$\text{Box}\left(\bigoplus_{i=1}^m S_i\right) = \bigoplus_{i=1}^m \text{Box}(S_i). \tag{7}$$

Property 5 [40]: For the r-order zonotope $\mathbb{Z} \in \mathbb{R}^n, \forall x = [x_1 \cdots x_n]^T \in \mathbb{Z}$, we can get

$$p_i - \sum_{j=1}^r |H_{ij}| \le x_i \le p_i + \sum_{j=1}^r |H_{ij}|, (i = 1, \ldots, n) \tag{8}$$

where H_{ij} is the ijth entry of the matrix H and $p_i \in \mathbb{R}^n$ is the center of the zonotope $\mathbb{Z}$.

In this paper, we consider a T-S fuzzy system described by the following fuzzy IF-THEN rules.

Rule i: If ι_{1k} is $\mathcal{G}_{i1}$ and ι_{2k} is $\mathcal{G}_{i2}$ and ... and $\iota_{\Lambda k}$ is $\mathcal{G}_{i\Lambda}$ then

$$\begin{cases} x_{k+1} = A_i x_k + B_i u_k + E_i w_k + F_i a_k^u, \\ y_k = C x_k + D a_k^y, \end{cases} \tag{9}$$

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where $i = 1, \ldots \mu$. The variables $x_k \in \mathbb{R}^n$,$u_k \in \mathbb{R}^{m_1}$,$y_k \in \mathbb{R}^p$, $w_k \in \mathbb{R}^{m_2}$, $a_k^u \in \mathbb{R}^{m_1}$ and $a_k^y \in \mathbb{R}^q$ stand for the system state, control input, system output, disturbance, actuator attacks and sensor attacks, respectively. For each ith subsystem, A_i, B_i, E_i and F_i represent the system matrix, input matrix, disturbance matrix and actuator attack matrix, respectively. The $(j = 1, \ldots, \Lambda)$ are the premise variables and $\iota_k = [\iota_{1k}, \ldots, \iota_{\Lambda k}]$ is the premise variable vector. $\mathcal{G}_{i1}$ to $\mathcal{G}_{i\Lambda}$ are fuzzy sets, and μ denote the whole number of subsystems and fuzzy rules. The membership functions $\tau_{ij}(\iota_{jk})$ can determine the membership ι_{jk} in fuzzy set $\mathcal{G}_{ij}$. Utilizing the center-average defuzzifier method, the T-S fuzzy system (9) can be transformed into

$$\begin{cases} x_{k+1} &= A(\sigma_k)x_k + B(\sigma_k)u_k + E(\sigma_k)w_k \\ & \quad +F(\sigma_k)a_k^u, \\ y_k &= Cx_k + Da_k^y, \end{cases} \tag{10}$$

where $A(\sigma_k) = \sum_{i=1}^{\mu} \sigma_i(\iota_k)A_i$, $B(\sigma_k) = \sum_{i=1}^{\mu} \sigma_i(\iota_k)B_i$, $E(\sigma_k) = \sum_{i=1}^{\mu} \sigma_i(\iota_k)E_i$, $F(\sigma_k) = \sum_{i=1}^{\mu} \sigma_i(\iota_k)F_i$ and $\sigma_i(\iota_k) = \frac{\prod_{j=1}^{\Lambda} \tau_{ij}(\iota_{jk})}{\sum_{i=1}^{\mu} \prod_{j=1}^{\Lambda} \tau_{ij}(\iota_{jk}))}$ are defined as the fuzzy weights, which satisfy $0 \leq \sigma_i(\iota_k) \leq 1$, $\sum_{i=1}^{\mu} \sigma_i(\iota_k) = 1$, $\forall i \in \{1, \ldots, \mu\}$.

Motivated by [27], w_k is split into two parts as follows

$$E(\sigma_k)w_k = \overline{E}_a M(\sigma_k)w_k^a + \overline{E}_b(\sigma_k)w_k^b, \tag{11}$$

where $w_k^a \in \mathbb{R}^{m_2}$ and $w_k^b \in \boldsymbol{W}_2 = \langle 0, H_{w2} \rangle$ are the components of disturbances to be decoupled and not be decoupled, respectively. $M(\sigma_k)$, $\overline{E}_b(\sigma_k)$ and $\overline{E}_a$ are the relevant parameterized matrices.

Assumption 1: We assume that the initial system state x_0 and the disturbance w_k are restricted to their respective zonotopes, which are represented as

$$x_0 \in \boldsymbol{X} = \langle p_0, H_0 \rangle, \quad w_k \in \boldsymbol{W} = \langle 0, H_w \rangle, \tag{12}$$

where $p_0 \in \mathbb{R}^n$, $H_0 \in \mathbb{R}^{n \times n}$ and $H_w \in \mathbb{R}^{m_2 \times m_2}$ are known vector and matrices.

Remark 1: Assumption 1 provides the basic uncertainty description required for the subsequent zonotopic reachability analysis and set-membership estimation. This is a common assumption in zonotope-based and interval-observer-based studies [40], [41]. In practical CPSs applications, these bounds can be estimated from sensor specifications, actuator tolerances, calibration tests, identification experiments, or historical data, and are usually selected with a conservative safety margin. If the actual uncertainty exceeds the prescribed zonotope, the set-inclusion and residual-bound guarantees may be weakened, thereby affecting the detection and reconstruction performance.

Assumption 2: Each matrix pair (A_i, C) of the fuzzy system (9) is detectable. The matrix C and the parameterized decoupled disturbance matrix $\bar{E}_a$ are required to satisfy $rank(C\bar{E}_a) = rank(\bar{E}_a) = m_2$.

Remark 2: To ensure that the system state x_k can be estimated from the measurement output y_k, the matrix pair (A_i, C) is required to be detectable. The rank condition $\text{rank}(C\overline{E}_a) = \text{rank}(\overline{E}_a) = m_2$ is imposed to guarantee the feasibility of disturbance decoupling in the observer design. Physically, this means that the disturbance directions to be decoupled should remain visible from the measured outputs, and thus sensors should be placed to provide independent information about these disturbance-affected state channels. Therefore, at least m_2 independent measurement directions are required in the sensor configuration. Similar assumptions are commonly adopted in existing unknown-input-observer and set-membership estimation studies, such as [42] and [43].

Assumption 3: To ensure that the actuator and sensor attacks can be reliably and uniquely reconstructed, it is assumed that the attack distribution matrices possess full column rank, i.e., $\text{rank}(F_i) = m_1$ for all $i \in \{1, \ldots, \mu\}$ and $\text{rank}(D) = q$.

Remark 3: This full column rank condition is a standard structural prerequisite in observer-based unknown input and attack reconstruction frameworks. It essentially ensures the existence of a unique mathematical solution for the pseudo-inverse operations without causing ambiguity, as widely adopted in recent state-of-the-art secure estimation literatures, such as [14], [44], and [45].

Algorithm 1 Overall Procedure of the Proposed Framework

Phase 1: Offline Observer Design (Section III-A)
Specify the system matrices and UBB noise bounds W_2 for the T-S fuzzy system (11).
Solve the LMI optimization problem in Theorem 1 to obtain P, Y, W_i.
Compute the observer gains T, N, and L_i using equations (18), (19), and (24).
Phase 2: Online Interval State Estimation and Attack Detection (Sections III-B & III-C)
for each time step $k \geq 0$ **do**
  Receive y_k, u_k, and compute fuzzy weights σ_k.
  Compute state estimate $\hat{x}_k$ using the T-N-L observer (13).
  Calculate interval bounds $[\underline{x}_k, \overline{x}_k]$ via Theorem 2.
  Execute **Algorithm 2** to generate thresholds $[\underline{\epsilon}_k, \overline{\epsilon}_k]$ via Theorem 3.
  if $\epsilon_k \notin [\underline{\epsilon}_k, \overline{\epsilon}_k]$ **then**
    Trigger attack alarm. Proceed to Phase 3.
  else
    System is attack-free. Return to Step 6.
  end if
end for
Phase 3: Attack Reconstruction and Isolation (Sections III-D & III-E)
Calculate zonotopic bounds $e_{k_d} \in \langle p_k^e, H_k^e \rangle$ using equation (48).
Execute **Algorithm 3** to reconstruct $\hat{a}_k^u$ and $\hat{a}_k^y$ via Theorem 4.
Isolate compromised channels $\hat{\mathcal{A}}_k^u$ and $\hat{\mathcal{A}}_k^y$ via Theorem 5.

III. Main Results

In this section, a comprehensive framework for interval state estimation, attack detection and reconstruction in T-S fuzzy systems is proposed. To clarify the logical interconnections among the following subsections, the overall execution procedure is summarized in the following Algorithm 1.

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A. The Design of H_∞ T-N-L Observer

In this part, we propose an improved T-N-L observer to estimate the system state. Inspired by the T-N-L observer proposed in [27], [30], [46], and [47], one can obtain the following observer:

$$\begin{aligned}\hat{x}_k = {} & TA(\sigma_{k-1})\hat{x}_{k-1} + TB(\sigma_{k-1})u_{k-1} \\ & + L(\sigma_{k-1})(y_{k-1} - C\hat{x}_{k-1}) + Ny_k, \end{aligned} \tag{13}$$

where $\hat{x}_k$ denotes the state estimation. The matrix $L(\sigma_k)$ is the T-N-L observer gain, which is defined as $L(\sigma_k) = \sum_{i=1}^{\mu} \sigma_i(\iota_k) L_i \in \mathbb{R}^{n\times p}$.

Then, the constant matrices $T \in \mathbb{R}^{n\times n}$ and $N \in \mathbb{R}^{n\times p}$ should be designed to satisfy

$$T + NC = I_n. \tag{14}$$

Accordingly, we define the error system e_k as $e_k = x_k - \hat{x}_k$. Based on (10), (13) and (14), we can obtain the following error system:

$$\begin{aligned} e_k = {} & [TA(\sigma_{k-1}) - LC]e_{k-1} + TF(\sigma_{k-1})a^u_{k-1} - NDa^y_k \\ & - LDa^y_{k-1} + T\overline{E}_a M(\sigma_{k-1})w^a_{k-1} \\ & + T\overline{E}_b(\sigma_{k-1})w^b_{k-1}. \end{aligned} \tag{15}$$

In order to reduce the influence of disturbances on the error system, the following condition should be satisfied:

$$T\overline{E}_a = 0. \tag{16}$$

According to (14) and (16), we can obtain the following equation:

$$\begin{bmatrix} T & N \end{bmatrix} \begin{bmatrix} I_n & \overline{E}_a \\ C & \mathbf{0} \end{bmatrix} = \begin{bmatrix} I_n & \mathbf{0} \end{bmatrix}. \tag{17}$$

Based on Lemma 1, we can get the matrices T and N as

$$T = \Gamma\Theta^\dagger\Lambda_1 + S_0\Psi\Lambda_1, \tag{18}$$

$$N = \Gamma\Theta^\dagger\Lambda_2 + S_0\Psi\Lambda_2, \tag{19}$$

where $\Gamma = [I_n \quad \mathbf{0}]$, $\Theta = \begin{bmatrix} I_n & \overline{E}_a \\ C & \mathbf{0} \end{bmatrix}$, $\Psi = I_n - \Theta\Theta^\dagger$, $\Lambda_1 = \begin{bmatrix} I_n \\ \mathbf{0} \end{bmatrix}$, $\Lambda_2 = \begin{bmatrix} \mathbf{0} \\ I_p \end{bmatrix}$ and S_0 is an arbitrary matrix.

Remark 4: The observer structure in (13) is motivated by [27], [30], [46], and [47]. Specifically, [30] originally proposed the T-N-L observer for linear time-varying descriptor systems, while [46] introduced the disturbance decomposition strategy in (11) for LPV systems. The works [27] and [47] further extended the T-N-L observer to discrete-time LPV descriptor systems and T-S fuzzy systems via zonotopic analysis, respectively. Different from these works, the proposed observer incorporates a disturbance decoupling condition (16) together with an H_∞ performance criterion, where the free matrix S_0 is analytically determined as $S_0 = P^{-1}Y$ through the LMI optimization in Theorem 1, yielding strictly tighter interval estimation compared to [27], [30], and [47].

Remark 5: In the process of designing T-N-L observer, the decoupling process can reduce the influence of disturbances on the error system e_k, which can also be regarded as an improved approach for designing T-N-L observer when disturbances can be partially decoupled. Furthermore, this method of T-N-L observer design can also be used in discrete-time linear systems, linear variable parameter systems and so on, which can greatly reduce the influence of disturbances on error systems e_k.

Consequently, by substituting (16) into (15), one can derive

$$\begin{aligned} e_{k_d} = {} & [TA(\sigma_{k-1}) - L(\sigma_{k-1})C]e_{k-1_d} + TF(\sigma_{k-1})a^u_{k-1} \\ & - NDa^y_k - L(\sigma_{k-1})Da^y_{k-1} \\ & + T\overline{E}_b(\sigma_{k-1})w^b_{k-1}. \end{aligned} \tag{20}$$

where e_{k_d} denotes the error system after decoupling process. By the definition of the error system, the system states can be obtained as

$$x_k = \hat{x}_k + e_{k_d}. \tag{21}$$

Subsequently, to forecast the bounds of the error system e_{k_d}, according to (21), the interval estimation of the system state can be derived as

$$\begin{cases} \bar{x}_k = \hat{x}_k + \bar{e}_k, \\ \underline{x}_k = \hat{x}_k + \underline{e}_k, \end{cases} \tag{22}$$

where $\overline{x}_k$ and $\underline{x}_k$ are the upper and lower bounds of the system state, respectively. Similarly, $\overline{e}_{k_d}$ and $\underline{e}_{k_d}$ denote the upper and lower bounds of the error system (20), respectively.

Then, the robust H_∞ T-N-L observer can be acquired by the following Theorem 1.

Theorem 1: Under the assumption that $a^u_k = 0$ and $a^y_k = 0$, given a scalar $\gamma > 0$, if there exists a positive definite matrix $P \in \mathbb{R}^{n\times n}(P \succ 0)$, matrices $W_i \in \mathbb{R}^{n\times p}(i = 1, \ldots, \mu)$ and $Y \in \mathbb{R}^{n\times p}$ satisfying the following condition:

$$\begin{bmatrix} I_n - P & * & * \\ 0 & -\gamma^2 I_{m_2} & * \\ \xi_1 & \xi_2 & -P \end{bmatrix} \prec 0, \tag{23}$$

where

$$\begin{aligned} \xi_1 &= P\Gamma\Theta^\dagger\Lambda_1 A_i + Y\Psi\Lambda_1 A_i - W_i C, \\ \xi_2 &= P\Gamma\Theta^\dagger\Lambda_1\overline{E}_{b,i} + Y\Psi\Lambda_1\overline{E}_{b,i}, \end{aligned}$$

and T, N and L_i can be obtained by

$$\begin{aligned} T &= \Gamma\Theta^\dagger\Lambda_1 + S_0\Psi\Lambda_1, \\ N &= \Gamma\Theta^\dagger\Lambda_2 + S_0\Psi\Lambda_2, \\ L_i &= P^{-1}W_i, \end{aligned}$$

where

$$S_0 = P^{-1}Y,$$

then (13) is a robust observer of the system (10) and $\|e_k\| < \gamma\|d_k\|$ is satisfied.

Proof: The Lyapunov function is defined as

$$V(k) = e^T_{k_d} P e_{k_d}. \tag{24}$$

Then, the criterion function is chosen as

$$J = \sum_{k=0}^{\infty} \left(\Delta V(k) + e^T_{k_d} e_{k_d} - \gamma^2 {w^b_k}^T w^b_k\right), \tag{25}$$

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where $\Delta V(k) = V(k+1) - V(k)$. In order to achieve the expected performance, we require $J < 0$. Then, one can obtain

$$\Delta V(k) + e_{k_d}^T e_{k_d} - \gamma^2 {w_k^b}^T w_k^b < 0. \tag{26}$$

The above condition (26) is the same as the following condition:

$$\begin{bmatrix} \Omega_{11} & * \\ \Omega_{21} & \Omega_{22} \end{bmatrix} \prec 0, \tag{27}$$

where

$$\begin{aligned} \Omega_{11} &= A_{\zeta,i}^T P A_{\zeta,i} - P + I_n, \\ \Omega_{21} &= (T\overline{E}_{b,i})^T P A_{\zeta,i}, \\ \Omega_{22} &= (T\overline{E}_{b,i})^T P T\overline{E}_{b,i} - \gamma^2 I_p, \\ A_{\zeta,i} &= TA_i - L_i C. \end{aligned}$$

According to the Schur complement lemma, we have

$$\begin{bmatrix} I_n - P & * & * \\ 0 & \gamma^2 I_{m_2} & * \\ \xi_1 & \xi_2 & -P \end{bmatrix} \prec 0,$$

where $\xi_1 = PA_{\zeta,i}$ and $\xi_2 = PT\overline{E}_{b,i}$. Based on the Schur complement lemma, the matrices T and N can be obtained by (18) and (19), respectively. Ultimately, following the bounded real lemma [48], (23) guarantees the stability of the error system (20) and the condition of $\|e_{k_d}\| < \gamma \|f_k\|$ is satisfied. □

Remark 6: The H_∞ performance of the proposed T-N-L observer is optimized by solving the optimization problem as follows

$$\min \quad \gamma^2, \quad \text{s.t. (23).} \tag{28}$$

Theorem 1 proposed a method to define the variable $S_0 = P^{-1}Y$, which avoids the need for manual selection of matrix S_0. This strategy improves the accuracy in determining matrices T, N and L_i, while concurrently enhancing the robust of the error system e_{k_d} and optimizing the computational efficiency.

B. State Interval Estimation Based on Zonotopic Analysis

Then, after designing the T-N-L observer by solving the optimization problem (28), the interval estimation of x_k can be obtained using the following Theorem 2, which relies on the reachability analysis of the error system (20).

Theorem 2: For the attack-free system, the interval estimation of the system state $[\underline{x}_k, \overline{x}_k]$ can be gained from (22) with $\hat{x}_0 = p_0$ and the interval estimation of error system e_{k_d}. The interval estimation of the initial error system e_{0_d} is represented as

$$[\overline{e}_{0_d}, \underline{e}_{0_d}] = \text{Box}(\langle 0, H_0 \rangle). \tag{29}$$

When $k = 0$, the interval estimation of the error system e_{1_d} is

$$\begin{aligned} [\overline{e}_{1_d}, \underline{e}_{1_d}] = &\text{Box}\left(A_\zeta(\sigma_0)\langle 0, H_0\rangle\right) \\ &\oplus \text{Box}\left(\overline{E}_b(\sigma_0)\mathbf{W}_2\right). \end{aligned} \tag{30}$$

When $k \geq 1$, the interval estimation of the error system e_{k_d} is

$$\begin{aligned} [\overline{e}_{k_d}, \underline{e}_{k_d}] = &\text{Box}\left(A_\zeta(\sigma_k)A_\zeta(\sigma_{k-1})\dots A_\zeta(\sigma_0)\langle 0, H_0\rangle\right) \\ &\oplus \bigoplus_{a=0}^{k-1} \text{Box}\left(A_\zeta(\sigma_k)\dots A_\zeta(\sigma_0)T\overline{E}_b(\sigma_a)\mathbf{W}_2\right) \\ &\oplus \text{Box}\left(T\overline{E}_b(\sigma_k)\mathbf{W}_2\right). \end{aligned} \tag{31}$$

Proof: For the error system (20), when $k = 0$, we can obtain the iteration equation of e_{1_d} as

$$e_{1_d} = A_\zeta(\sigma_0)e_{0_d} + \overline{E}_b(\sigma_0)w_0^b. \tag{32}$$

When $k \geq 1$, we can derive

$$\begin{aligned} e_{k_d} = &A_\zeta(\sigma_k)A_\zeta(\sigma_{k-1})\dots A_\zeta(\sigma_0)e_{0_d} \\ &- \sum_{i=0}^{k-2} A_\zeta(\sigma_{k-1})\dots A_\zeta(\sigma_0)T\overline{E}_b(\sigma_a)w_{k-1-i}^b \\ &+ T\overline{E}_b(\sigma_{k-1})w_{k-1}^b. \end{aligned} \tag{33}$$

The reachability analysis of the error system e_{k_d} is represented as $\varkappa_k^e$. According to (32), (33) and definition 3, when $k = 0$, one can get

$$\text{Box}(\varkappa_1^e) = \text{Box}(A_\zeta(\sigma_0)\varkappa_0^e) \oplus \text{Box}(\overline{E}_b(\sigma_0)\mathbf{W}_2). \tag{34}$$

When $k \geq 1$, one can gain

$$\begin{aligned} \text{Box}\left(\varkappa_k^e\right) = &\text{Box}\left(A_\zeta(\sigma_k)A_\zeta(\sigma_{k-1})\dots A_\zeta(\sigma_0)\langle 0, H_0\rangle\right) \\ &\oplus \bigoplus_{a=0}^{k-1} \text{Box}\left(A_\zeta(\sigma_k)\dots A_\zeta(\sigma_0)T\overline{E}_b(\sigma_a)\mathbf{W}_2\right) \\ &\oplus \text{Box}\left(T\overline{E}_b(\sigma_k)\mathbf{W}_2\right). \end{aligned} \tag{35}$$

Owing to the initial system state $x_0 \in \langle p_0, H_0\rangle$ and $\hat{x} = p_0$, we have $e_{0_d} = x_0 - \hat{x}_0 \in \langle 0, H_0\rangle$. Therefore, one can obtain

$$\text{Box}(\varkappa_1^e) = \text{Box}\left(\langle 0, H_0\rangle\right). \tag{36}$$

In conclusion, $\underline{e}_{k_d}$ and $\overline{e}_{k_d}$ efficiently computed based on above discussion. Then, according to (22), we can derive the upper and lower bounds of the system state x_k.

C. Attack Detection Based on Zonotopic Analysis

In this subsection, an improved attack detection mechanism for T-S fuzzy systems is proposed to overcome the limitations of conservative thresholds. By leveraging zonotopic reachability analysis, tighter interval bounds for the system residual ε_k are dynamically computed at each time step. This allows for highly sensitive attack detection by simply identifying when the real-time residual breaches these bounds, effectively preventing false alarms caused by UBB disturbances. The detailed formulation is presented in Theorem 3.

Theorem 3: Under attack-free condition, the interval estimation of the residual can be obtained as $[\underline{\varepsilon}_k, \overline{\varepsilon}_k]$ can be gained. The initial interval estimation of the residual ε_0 can be obtained as

$$[\overline{\varepsilon}_0, \underline{\varepsilon}_0] = \text{Box}\left(\langle 0, CH_0\rangle\right). \tag{37}$$

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When $k=0$, the interval estimation of the residual ε_1 can be represented as

$$[\overline{\varepsilon}_1, \underline{\varepsilon}_1] = \text{Box}\left(CA_\zeta(\sigma_0)\langle 0, H_0\rangle\right) \oplus \text{Box}\left(C\overline{E}_b(\sigma_0)\mathbf{W}_2\right). \tag{38}$$

When $k \geq 1$, the interval estimation of residual ε_{k+1} cam be gained as

$$\begin{aligned}\left[\overline{\varepsilon}_{k+1}, \underline{\varepsilon}_{k+1}\right] &= \text{Box}\left(CA_\zeta(\sigma_k)A_\zeta(\sigma_{k-1})\ldots A_\zeta(\sigma_0)\langle 0, H_0\rangle\right) \\ &\oplus \bigoplus_{a=0}^{k-1} \text{Box}\left(CA_\zeta(\sigma_k)\ldots A_\zeta(\sigma_0)T\overline{E}_b(\sigma_a)\mathbf{W}_2\right) \\ &\oplus \text{Box}\left(CT\overline{E}_b(\sigma_k)\mathbf{W}_2\right).\end{aligned} \tag{39}$$

Proof: According to the definition of residual ε_k, the residue ε_k can be shown as

$$\begin{aligned}\varepsilon_k &= y_k - C\hat{x}_k \\ &= CA_\zeta(\sigma_{k-1})e_{k-1_d} + CT\overline{E}_b(\sigma_{k-1})w^b_{k-1}.\end{aligned} \tag{40}$$

So the initial value of the residual is shown as

$$\varepsilon_0 = Ce_{0_d}. \tag{41}$$

When $k=0$, the residual ε_1 can be expressed as

$$\varepsilon_1 = CA_\zeta(\sigma_0)e_{0_d} + CT\overline{E}_b(\sigma_0)w^b_0. \tag{42}$$

When $k \geq 1$, the residual ε_k can be obtained as

$$\begin{aligned}\varepsilon_k &= CA_\zeta(\sigma_k)A_\zeta(\sigma_{k-1})\ldots A_\zeta(\sigma_0)e_{0_d} \\ &- \sum_{i=0}^{k-2} CA_\zeta(\sigma_k)\ldots A_\zeta(\sigma_0)T\overline{E}_b(\sigma_a)w^b_{k-1-i} \\ &+ CT\overline{E}_b(\sigma_{k-1})w^b_{k-1}.\end{aligned} \tag{43}$$

The reachability set of the residual ε_k is defined as $\varkappa^\varepsilon_k$. Based on $x_0 \in \langle p_0, H_0\rangle$ and $\hat{x}_0 = p_0$, we can get $e_0 = x_0 - \hat{x}_0 \in \langle 0, H^a_0\rangle$. Then, the initial residual $\varkappa^\varepsilon_0$ can be obtained as

$$\text{Box}\left(\varkappa^\varepsilon_0\right) = \text{Box}\left(\langle 0, CH_0\rangle\right). \tag{44}$$

When $k=0$, $\varkappa^\varepsilon_1$ can be described as

$$\text{Box}\left(\varkappa^\varepsilon_1\right) = \text{Box}\left(CA_\zeta(\sigma_0)\langle 0, H_0\rangle\right) \oplus \text{Box}\left(C\overline{E}_b(\sigma_0)\mathbf{W}_2\right). \tag{45}$$

When $k \geq 1$, $\varkappa^\varepsilon_k$ can be achieved as

$$\begin{aligned}\text{Box}\left(\varkappa^\varepsilon_k\right) &= \text{Box}\left(CA_\zeta(\sigma_k)A_\zeta(\sigma_{k-1})\ldots A_\zeta(\sigma_0)\langle 0, H_0\rangle\right) \\ &\oplus \bigoplus_{a=0}^{k-1} \text{Box}\left(CA_\zeta(\sigma_k)\ldots A_\zeta(\sigma_0)T\overline{E}_b(\sigma_a)\mathbf{W}_2\right) \\ &\oplus \text{Box}\left(CT\overline{E}_b(\sigma_k)\mathbf{W}_2\right).\end{aligned} \tag{46}$$

Subsequently, the interval estimation of the residual ε_k is obtained such that $[\overline{\varepsilon}_k, \underline{\varepsilon}_k] = \text{Box}\left(\varkappa^\varepsilon_k\right)$. When the system is attack-free, $\overline{\varepsilon}_k < \varepsilon_k < \underline{\varepsilon}_k$ is satisfied. □

Based on the above discussion, we propose Algorithm 2 to calculate the interval estimate of the system residual ε_k such that $[\overline{\varepsilon}_k, \underline{\varepsilon}_k]$. If the residual exceeds the boundary, an attack will be detected. The proposed attack detection method based on the residual is defined as

$$\begin{cases}\varepsilon_k \in [\underline{\varepsilon}_k, \overline{\varepsilon}_k], & \text{attack-free;} \\ \varepsilon_k \notin [\underline{\varepsilon}_k, \overline{\varepsilon}_k], & \text{attacked.}\end{cases} \tag{47}$$

Remark 7: The proposed method achieves high computational efficiency through an iterative linear-complexity algorithm, compact zonotope representations, and a decoupling strategy that reduces conservatism. Unlike [31] and [49], it eliminates cooperative constraints while maintaining accuracy for real-time applications, and replaces complex geometric checks with efficient interval arithmetic for attack detection.

D. Attack Reconstruction Based on Zonotopic Analysis

In this subsection, based on the estimation of the system state x_k computed by the proposed T-N-L observer, FDI attacks can be estimated. According to the definition of the error system, the system state x_k can be obtained as $x_k = e_k + \hat{x}_k$. Based on (20), we can confine the error system e_{k_d} to their zonotopes as follows

$$e_{k_d} \in \langle p^e_k, H^e_k\rangle, \tag{48}$$

where

$$\begin{aligned}p^e_k &= A_\zeta(\sigma_{k-1})e_{k-1_d} + TF(\sigma_{k-1})a^u_{k-1} - NDa^y_{k-1} \\ &- L(\sigma_{k-1})Da^y_{k-1}, \\ H^e_k &= \left[A_\zeta(\sigma_{k-1}) \downarrow_q \left(H^e_{k-1}\right) \quad T\overline{E}_b(\sigma_{k-1})H_{w2}\right].\end{aligned}$$

Then, the reconstruction of actuator attacks and sensor attacks will be given in the following Theorem 4.

Theorem 4: Consider the system (10) under FDI attacks. Then the actuator attacks and sensor attacks can be reconstructed as

$$\hat{a}^u_k = F(\sigma_k)^\dagger(\Xi_k - A(\sigma_{k-1})\Xi_{k-1} - B(\sigma_{k-1})u_{k-1}), \tag{49}$$

$$\hat{a}^y_k = D^\dagger(y_k - C\Xi_k), \tag{50}$$

where

$$\hat{a}^u_k \in \langle p_{au}, H_{au}\rangle, \quad \hat{a}^y_k \in \langle p_{ay}, H_{ay}\rangle,$$

with

$$\begin{aligned}p_{au} &= F(\sigma_k)^\dagger(\Xi_k - A(\sigma_{k-1})\Xi_{k-1} - B(\sigma_{k-1})u_{k-1}), \\ H_{au} &= F(\sigma_k)^\dagger[H^e_k \quad A(\sigma_{k-1})H^e_{k-1} \quad EH_w], \\ p_{ay} &= D^\dagger(y_k - C\Xi_k), \\ H_{ay} &= [D^\dagger CH^e_k], \quad \Xi_k = p^e_k + \hat{x}_k.\end{aligned}$$

Proof: Based on the definition of the error system and system model, we can derive

$$\begin{aligned}e_{k+1} + \hat{x}_{k+1} &= A(\sigma_k)(e_k + \hat{x}_k) + B(\sigma_k)u_k + F(\sigma_k)a^u_k \\ &+ E(\sigma_k)w_k.\end{aligned}$$

Then, one can calculate

$$\begin{aligned}a^u_k &= F(\sigma_k)^\dagger[e_{k+1} - A(\sigma_k)e_k + \hat{x}_{k+1} - A(\sigma_k)\hat{x}_k \\ &-E(\sigma_k)w_k - B(\sigma_k)u_k].\end{aligned} \tag{51}$$

Similarly, from the system (10) in the present of FDI attacks, we can obtain

$$y_k = C(e_k + \hat{x}_k) + Da^y_k. \tag{52}$$

Therefore, one can get

$$a^y_k = D^\dagger(y_k - C(e_k + \hat{x}_k)). \tag{53}$$

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Algorithm 2 Attack Detection Threshold Generation Based on the System Residual

Require: u_k, y_k
Ensure: $\underline{\varepsilon}_k, \overline{\varepsilon}_k$
1: **Initialization:**
2: $\hat{x}_0 = p_0, \mathbf{T}_x^0 = \langle 0, H_0 \rangle, \mathbf{T}_w^0 = \emptyset$
3: **for** $k \geq 0$ **do**
4: $[\underline{\varepsilon}_k, \overline{\varepsilon}_k] = \text{Box}(\mathbf{T}_x^k) \oplus \mathbf{T}_1 \oplus \mathbf{T}_w^k$
5: $\mathbf{T}_x^{k+1} = CA_\zeta(\sigma_k)\mathbf{T}_x^k$
6: $\mathbf{T}_1 = \text{Box}(C\overline{T}E_b(\sigma_k)\mathbf{W}_2)$
7: $\mathbf{T}_w^{k+1} = \mathbf{T}_w^k \oplus \text{Box}\left(CA_\zeta(\sigma_k)\overline{E}_b(\sigma_k)\mathbf{W}_2\right)$
8: **end for**

In practical systems, it is pointed out that the error system e_{k_d} and disturbances w_k cannot be precisely measured but can be limited in a zonotope, respectively. So one can obtain

$$\hat{a}_k^u = F(\sigma_k)^\dagger (p_k^a + \hat{x}_k - A(\sigma_{k-1})(p_{k-1}^a + \hat{x}_{k-1}) - B(\sigma_{k-1})u_{k-1}), \tag{54}$$

$$\hat{a}_k^y = D^\dagger (y_k - C(p_k^a + \hat{x}_k)). \tag{55}$$

Since the parameters y_k, $\hat{x}_k$, u_k and w_k can be directly obtained as zonotopes, respectively, one can derive

$$\hat{a}_k^u \in \langle p_{au}, H_{au} \rangle, \quad \hat{a}_k^y \in \langle p_{ay}, H_{ay} \rangle,$$

where the specific expressions of p_{au}, H_{au}, p_{ay} and H_{ay} are given in upon discussion. The reconstruction of actuator attack $\hat{a}_k^u$ and sensor attack $\hat{a}_k^y$ are selected based on (49) and (50), respectively. The proof is completed. □

Remark 8: Furthermore, regarding numerical robustness, if the attack distribution matrices $F(\sigma_k)$ and D are ill-conditioned, the pseudo-inverse operations may become numerically sensitive, potentially amplifying bounded disturbances. However, a fundamental advantage of the proposed zonotopic framework is that such amplified uncertainties are strictly encapsulated within the generated interval bounds (H_{au} and H_{ay}), following the guaranteed containment property of set-membership estimation [50], [51]. Consequently, while ill-conditioning may yield more conservative (wider) reconstruction bounds, the true attack signals are mathematically guaranteed to remain within these robust limits, a feature that has been increasingly leveraged in recent secure estimation studies to handle modeling uncertainties and numerical perturbations [36], [52].

Theorem 4 investigates the impact of disturbances and the error system (20) on the reconstruction of the attacks. All FDI attacks can be thoroughly reconstructed if the disturbances are not considered.

E. Attack Isolation Based on Zonotopic Analysis

In this subsection, an improved attack isolation method based on zonotopic analysis is proposed in Theorem 5.

Theorem 5: We assume that $\hat{\mathcal{A}}_{k,i}^u$ and $\hat{\mathcal{A}}_k^y$ represent the sets of isolated actuator and sensor attacks. The isolation of FDI attacks can be accomplished in accordance with

$$\hat{\mathcal{A}}_k^u = \text{supp}(\tau_{k,i}^u), \quad \hat{\mathcal{A}}_k^y = \text{supp}(\tau_{k,i}^y), \tag{56}$$

where $\tau_{k,i}^u = 0$ if $\hat{a}_{k,i}^u \in \langle 0, H_{au} \rangle$; otherwise, $\tau_{k,i}^u = 1$. $\tau_{k,i}^y = 0$ if $\hat{a}_{k,i}^y \in \langle 0, H_{ay} \rangle$; otherwise, $\tau_{k,i}^y = 1$.

Proof: According to Theorem 4, we can obtain $\hat{a}_k^u \in \langle p_{au}, H_{au} \rangle$ and $\hat{a}_k^y \in \langle p_{ay}, H_{ay} \rangle$. Then, if the i-th actuator is attack-free, we can get that $\hat{a}_{k,i}^u \in \langle 0, H_{au} \rangle$. On the other hand, if the i-th actuator is attacked, the condition needs to satisfy $\hat{a}_{k,i}^u \notin \langle 0, H_{au} \rangle$. Now, the isolation of the actuator attacks is represented as

$$\begin{cases} \tau_{k,i}^u = 0, & \text{attack-free;} \\ \tau_{k,i}^u = 1, & \text{attacked.} \end{cases}$$

Similarly, the isolation of the sensor attacks can be accomplished too. □

The reconstruction and isolation of FDI attacks utilising the proposed T-N-L observer can be completed in accordance with the following Algorithm 3.

Remark 9: While assuming accurately known plant matrices and fuzzy weights is standard in recent T-S fuzzy secure estimation literature (e.g., [2], [40], [47], and [53]), practical CPSs inevitably face modeling inaccuracies. Fortunately, our framework exhibits inherent robustness against minor bounded parametric uncertainties by treating them as pseudo-disturbances. These are effectively attenuated by the H_∞ performance index and safely encapsulated within the widened zonotopic bounds, preserving reliable state and attack estimation.

Remark 10: The proposed framework is closely related to the scope of IEEE Transactions on Green Communications and Networking (TGCN). Recent TGCN studies have addressed secure control under DoS/deception attacks, remote state estimation, and energy-aware transmission scheduling in networked CPSs [20], [21], [22], [23]. These works show that cyber-security, communication reliability, and resource efficiency are closely coupled in green CPSs. Different from these studies, this paper focuses on attack detection, reconstruction, and isolation for T-S fuzzy CPSs under UBB disturbances. The proposed zonotopic residual analysis and attack reconstruction provide useful diagnostic information for subsequent resilient control, thereby supporting reliable and resource-efficient CPSs operation.

IV. Simulation Study

In this section, we consider the following T-S fuzzy system in the presence of FDI attacks, and the system parameters are

$$A_1 = \begin{pmatrix} 0.3 & 0.1 & 0 \\ 0.2 & 0.5 & 0.1 \\ 0 & -0.5 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0.6 & 0.1 & 0 \\ 0 & 0.5 & 0.1 \\ 0 & -0.2 & 0 \end{pmatrix},$$

$$B_1 = B_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \overline{E}_{b1} = \begin{pmatrix} 0.2 \\ -0.3 \\ 0.2 \end{pmatrix}, \quad M_1 = M_2 = 1,$$

$$\overline{E}_{b2} = \begin{pmatrix} 0.3 \\ 0.2 \\ -0.1 \end{pmatrix}, \quad F_1 = F_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \overline{E}_a = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix}.$$

In this case, the disturbances and input are set to satisfy the following condition: $w_k^a = w_k^b = 0.5sin(0.2k)$ and

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Algorithm 3 Attack Reconstruction and Isolation

Require: A, B, C, p_k^e, y_k, and $\hat{x}_k$

1: **while** $k \leq k_N$ **do**

2: $\hat{a}_k^u = F(\sigma_k)^\dagger(p_k^a + \hat{x}_k - A(\sigma_{k-1})(p_{k-1}^a + \hat{x}_{k-1}) - B(\sigma_{k-1})u_{k-1})$

3: $\hat{a}_k^y = D^\dagger(y_k - C(p_k^a + \hat{x}_k))$

4: $\hat{\mathcal{A}}_k^u = \text{supp}(\tau_{k,i}^u)$

5: $\hat{\mathcal{A}}_k^y = \text{supp}(\tau_{k,i}^y)$

6: **end while**

Ensure: $\hat{a}_k^u$, $\hat{a}_k^y$, $\hat{\mathcal{A}}_k^u$, and $\hat{\mathcal{A}}_k^y$

$u_k = 2cos(0.1k)$, respectively. The initial system state is set to $x_0 = [0.5 \quad -0.5 \quad 0.5]^T$.

Then, the membership function is represented as

$$\sigma_1(x_1(k)) = \begin{cases} \dfrac{\sin(x_1(k))}{x_1(k)}, & x_1(k) \neq 0, \\ 1, & x_1(k) = 0, \end{cases}$$
$$\sigma_2(x_1(k)) = 1 - \sigma_1(x_1(k)),$$

We assume that the premise variable x_1 is measurable, whereas x_2 and x_3 are unmeasurable. Hence, we set $C = [1 \quad 0 \quad 0]$ and $D = 0.2$. Then, the values of all elements of the observer gains can be calculated by solving (21) as

$$T = \begin{pmatrix} -3.89 \times 10^{-16} & 1.94 \times 10^{-16} & -2.5 \times 10^{-16} \\ -0.7727 & 1 & 1.11 \times 10^{-16} \\ 1 & -2.22 \times 10^{-16} & 1 \end{pmatrix},$$
$$N = \begin{pmatrix} 1 \\ 0.7727 \\ -1 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 4.01 \times 10^{-12} \\ 0.65 \\ 0.3 \end{pmatrix},$$
$$L_2 = \begin{pmatrix} 4.24 \times 10^{-9} \\ -0.4 \\ 0.2 \end{pmatrix}.$$

Now, under the attack-free condition, the state estimation can be obtained by the H_∞ T-N-L observer in Theorem 1. Moreover, the reachability analysis of the error system e_{k_d} can be derived by Theorem 2. Then, based on (22), the interval estimation of the system state can be absolutely achieved.

To validate the effectiveness of attack detection, reconstruction, and isolation under simultaneous sensor and actuator attacks, we define these attacks as

$$a_k^u = \begin{cases} 5cos(0.1k_a), & k_a \in [15, 40]; \\ 0, & k_a \notin [15, 40]. \end{cases}$$
$$a_k^y = \begin{cases} 5sin(0.1k_a), & k_a \in [15, 40]; \\ 0, & k_a \notin [15, 40]. \end{cases}$$

To confirm generality, a standard two-rule T-S fuzzy benchmark is adopted, scaling disturbance and attack magnitudes to rigorously stress the observer. Crucially, integrating H_∞ optimization with zonotopic containment encapsulates amplitude variations within dynamic bounds, ensuring robustness across typical operating conditions. Under these simulation parameters' settings (Table II), quantitative performance is evaluated using root mean square error (RMSE) for estimation and reconstruction accuracy, and missed detection rate (MDR) for detection reliability, with comparative results summarized in Table III.

TABLE II
SIMULATION PARAMETERS

Parameters	Values
Total simulation steps (k_N)	99
Attack occurrence time (k_a)	$15 \sim 45$
Initial zonotope bound (H_0)	$\text{diag}(0.1, 0.1, 0.1)$
Disturbance zonotope bound (H_{w2})	0.5
Optimized H_∞ index (γ)	6.99×10^{-7}
Maximum zonotope order	10

TABLE III
QUANTITATIVE PERFORMANCE COMPARISON

Evaluation Metrics	Proposed Method	Existing Methods
RMSE of State Est. (x_1)	4.6898	4.6898
RMSE of State Est. (x_2)	4.0222	4.8013
RMSE of State Est. (x_3)	3.9645	4.2633
RMSE of Actuator Attack ($\hat{a}_k^u$)	0.2237	0.2362
RMSE of Sensor Attack ($\hat{a}_k^y$)	0.0027	0.0274
Missed Detection Rate (MDR)	0%	4.8387%

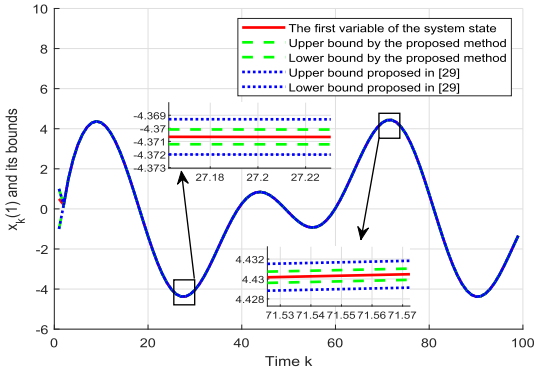


Fig. 2. The system state $x_k(1)$ and its interval estimation.

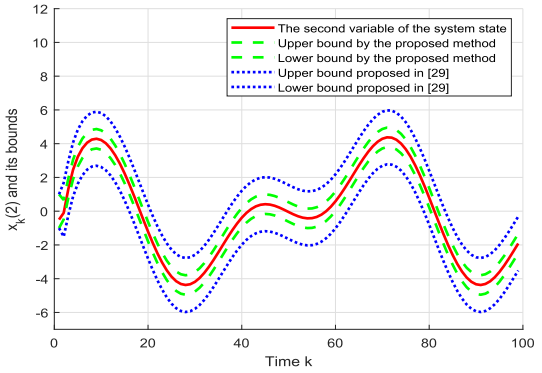


Fig. 3. The system state $x_k(2)$ and its interval estimation.

The simulation results are displayed in Figs. 2–9. Figs. 2–4 are generated by the proposed H_∞ T-N-L observer and reachability analysis. These figures compare the system state bounds calculated by our method with those obtained without the decoupling mechanism. As quantitatively verified in Table III, the state interval estimations generated by our

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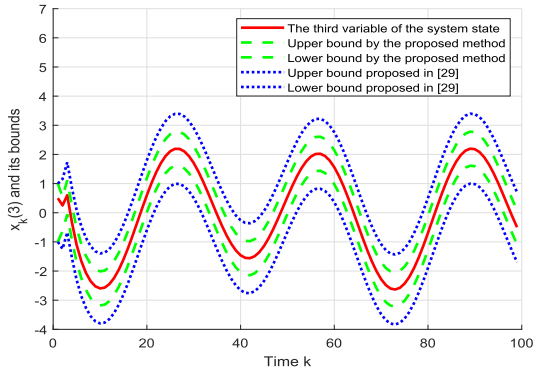


Fig. 4. The system state $x_k(3)$ and its interval estimation.

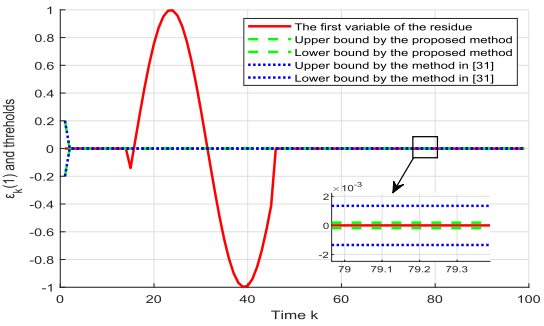


Fig. 5. The residue $\varkappa_k(1)$ and thresholds $\varkappa_k^{\varepsilon}$.

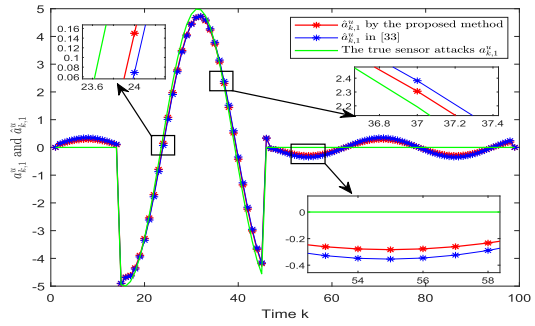


Fig. 6. The true actuator attack $a^u_{k,1}$ and its reconstructions.

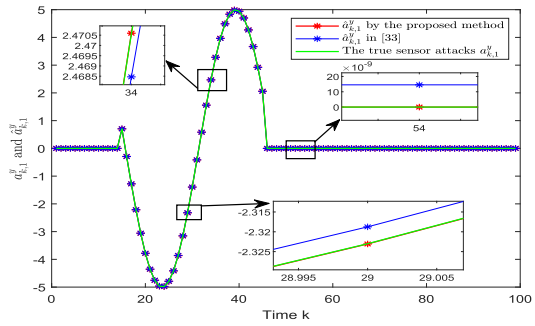


Fig. 7. The true sensor attack $a^y_{k,1}$ and its reconstructions.

proposed method exhibit significantly higher accuracy and reliability compared to traditional approaches [29], [30].

Fig. 5, obtained by applying Theorem 3, demonstrates the attack detection approach based on residue reachability analysis. It compares the residue bounds calculated using

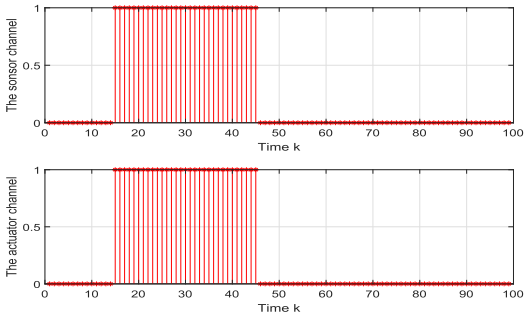


Fig. 8. Attack isolation of the proposed method.

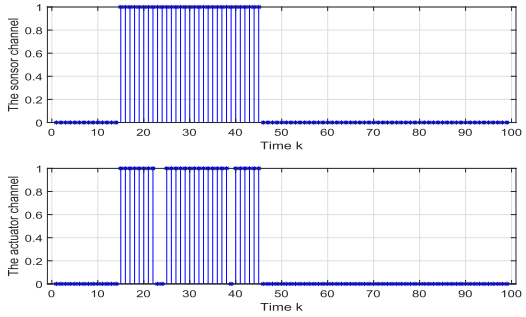


Fig. 9. Attack isolation of the traditional method.

our method with those lacking the decoupling mechanism. The results show that the residue strictly remains within the bounds when attack-free and promptly breaches them during an attack. Notably, our approach achieves a perfect 0% MDR, significantly outperforming existing methods [31], [32] (4.8387%).

The attack reconstruction results based on Theorem 4 are shown in Figs. 6–7. Fig. 6 represents the reconstruction of the actuator attack a^u_k, while Fig. 7 denotes the sensor attack a^y_k. Both figures show that the reconstructed attacks closely follow the actual signals. Crucially, our method significantly improves precision, achieving an approximate 90.1% RMSE reduction (from 0.0274 to 0.0027) for sensor attacks compared to the methods in [33].

The isolation of FDI attacks a^u_k and a^y_k is depicted in Figs. 8–9. Fig. 8 demonstrates that our method effectively isolates attacks on actuator and sensor components, clearly indicating the compromised time intervals. In contrast, Fig. 9 shows the inferior isolation performance using traditional method [16]. These results confirm our approach successfully isolates all attacks.

V. Limitation

Despite the promising results demonstrated above, several limitations of the proposed framework warrant further discussion. First, the proposed method is built upon Assumption 1, which requires the zonotopic bounds on the initial state and disturbances to be known a priori. However, such precise bound information may not always be readily available in practical applications, potentially compromising the tightness of the resulting interval estimation. Second, Assumption 2 imposes a rank condition, i.e., $\text{rank}(CE_a) = \text{rank}(E_a) = m_2$,

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which introduces structural constraints on the system matrices and may restrict the applicability of the proposed approach to certain classes of T-S fuzzy systems. Third, the current framework is developed exclusively for discrete-time T-S fuzzy systems, and its extension to continuous-time counterparts remains a challenging open problem that deserves further investigation. Finally, handling severe unmodeled nonlinearities or significant parameter shifts remains an open challenge. Absorbing excessively large modeling errors into the UBB disturbances can lead to overly conservative interval bounds, which may reduce the sensitivity of the attack detection mechanism. Balancing robustness and detection conservatism under severe modeling mismatches will be explored in future work.

VI. Conclusion

In this work, to represent the complex nonlinear systems in CPSs, a T-S fuzzy model is used to handle this problem. Based on the proposed T-N-L observer and reachability analysis for T-S fuzzy systems, a tighter interval estimation of the system states is obtained. Then, an improved attack detection approach for T-S fuzzy systems is proposed based on residual reachability analysis, demonstrating strong robustness against disturbances and enhanced sensitivity to attacks. Subsequently, the accuracy of malicious attack reconstruction and isolation is enhanced by the proposed method. Finally, the simulation results show the effectiveness and feasibility of the proposed approach. In future work, we will extend the proposed framework to more general networked CPSs scenarios involving richer attack patterns, distributed filtering, secure cooperative control, and fault-isolation mechanisms, inspired by recent studies on CPSs security and fault diagnosis [54], [55], [56]. Moreover, reinforcement learning-based methods will be considered to further enhance adaptive decision-making and attack-tolerant compensation in uncertain and dynamic CPSs environments, especially by enabling data-driven policy optimization for resilient operation under complex attack scenarios [57], [58], [59].

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